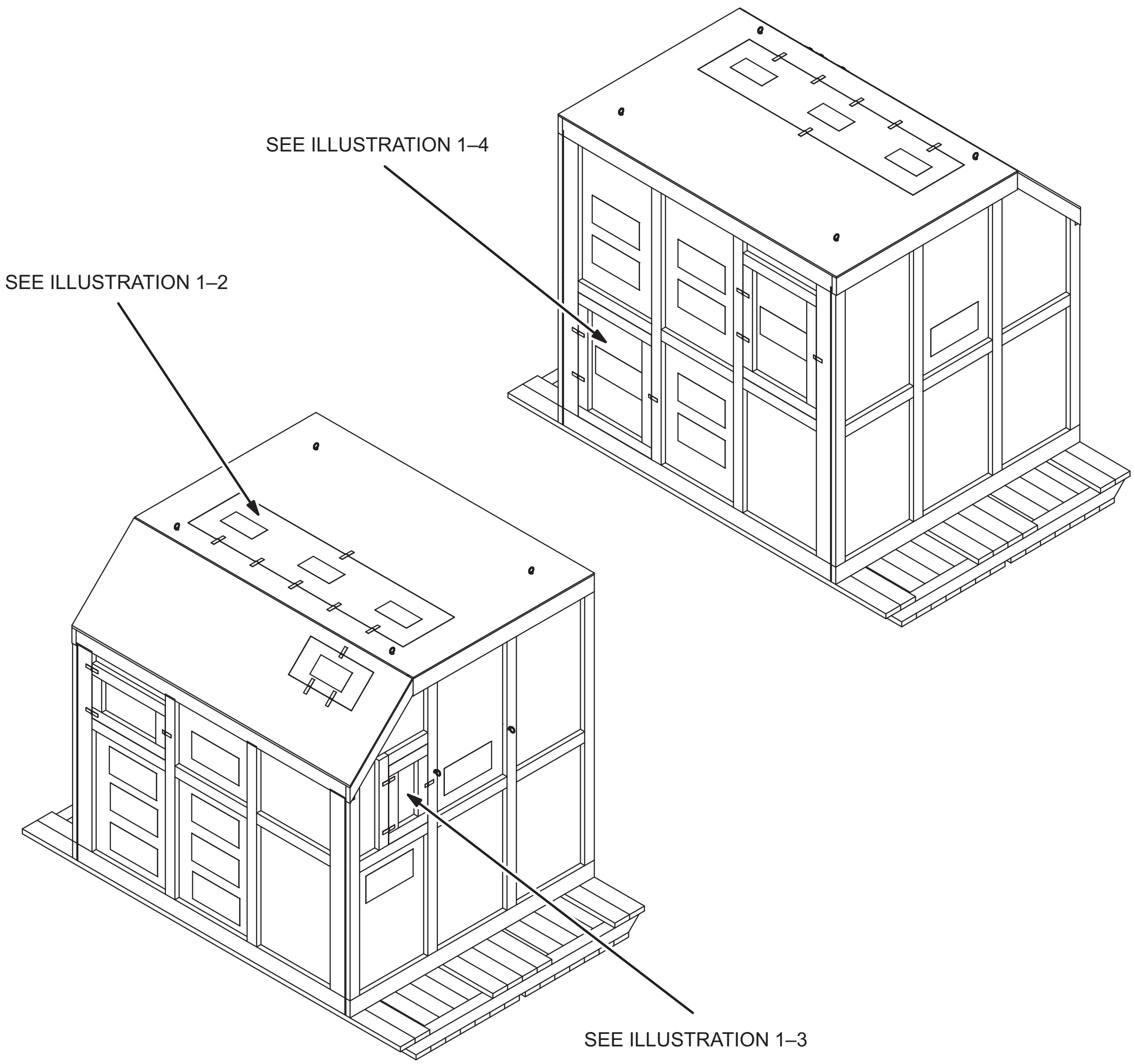
SECTION 1– SHIPPING / DELIVERY INSTRUCTIONS

1–1 INTRANSIT SERVICE

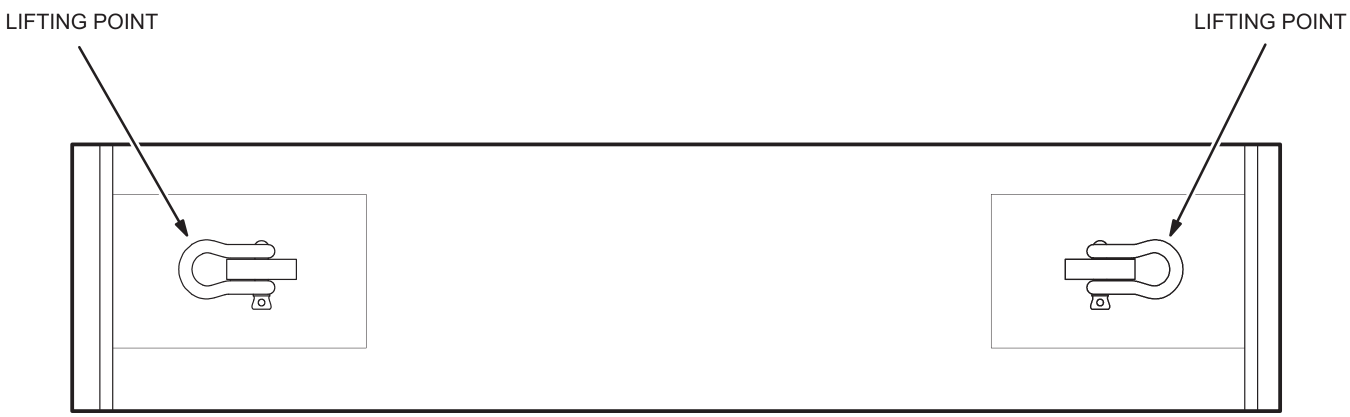
Note

The following information outlines provisions for filling and electrical / temperature checks while magnet is still in shipping crate. This will enable required service to be performed in transit based upon magnet shipping date. The following Illustrations (1–1 through 1–3) show the magnet configuration behind each shipping crate access door. Access doors and hinges not shown for clarity.



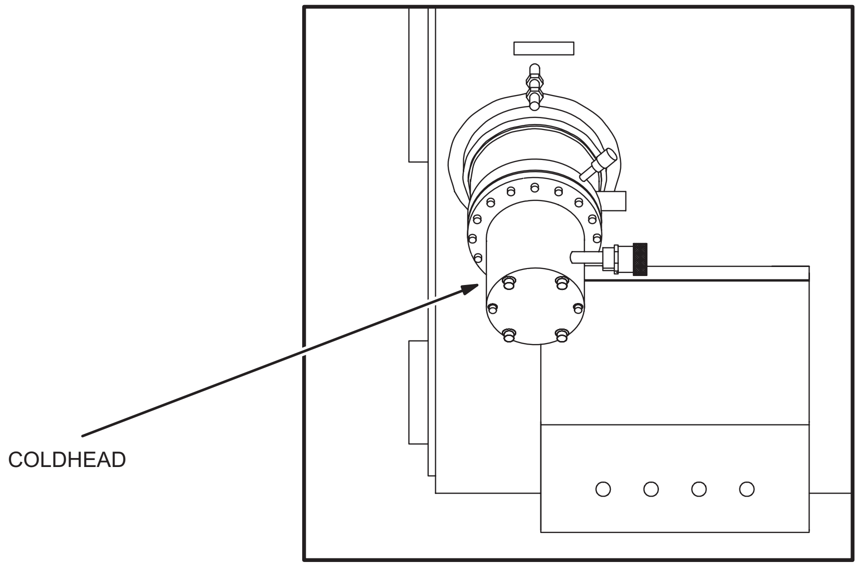
ACCESS DOOR LABELING ON CRATE
ILLUSTRATION 1–1

1–1 INTRANSIT SERVICE (continued)



TOP ACCESS DOOR LABELED "LIFTING POINT"
ILLUSTRATION 1–2

1–1 INTRANSIT SERVICE (continued)



LEFT SIDE ACCESS DOOR LABELED "COLD HEAD ACCESS"
ILLUSTRATION 1–3

1–2 SHIPPING / HANDLING

CAUTION

Shipping and handling guidelines are provided in Table 1–1.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

Review guidelines with Shipper / Riggers prior to transporting magnet.

TABLE 1–1
MAGNET SHIPPING AND HANDLING INFORMATION

MAGNET	* MAXIMUM WEIGHT	** MAXIMUM TILT	(1) ALLOWABLE SHIPPING MODES	(2) FORKLIFT CAPABILITY
SV	SV = 18,000 lbs.	30 deg.	A, T, Tr, B	Yes

MAGNET	SHIPPING CAPABILITY	(3) MAXIMUM TRANSIT TIME	MAXIMUM SHOCK LOADS	COMMENTS
SV	Cold	21 days	2 G’s	See Notes

* Includes weight of Shipping Crate (2,200 lbs.)

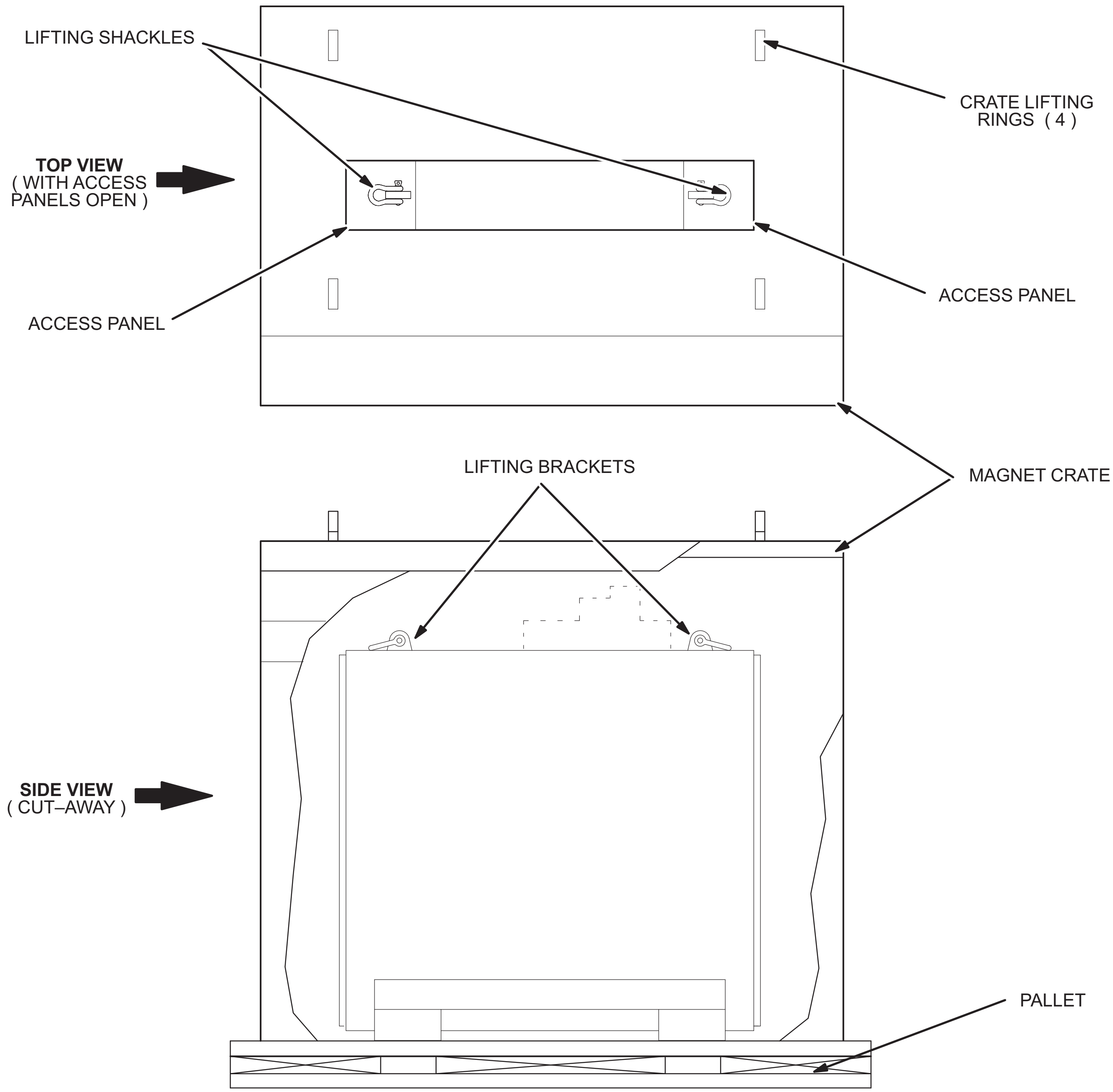
** Tilt allowed when suspended by Lifting Shackles.

Notes:

1. Key for Shipping mode symbols:
   “A” Airplane (including any plane that has fuselage openings large enough to accept a magnet)
   “T” Air ride Trailer (Any magnet transported on a non–air ride trailer must be identified and never used in a Mobile trailer.)
   “Tr” Train
   “B” Boat or ocean going ship
2. Extreme care must be exercised during forklift operations. The magnet pallet / crate must be picked up from the sides only. The forks MUST be placed directly under the four (4) feet of the magnet. The magnet feet can be identified by the steel plates attached to the pallet.
3. The elapsed time begins when the magnet leaves the Florence loading dock.

1–3 UNLOADING / LOADING MAGNET AND CRATE

Note
Lifting Shackles and Brackets are provided for magnet unloading / loading. See Illustration 1–4.
Shackle / Bracket access panels are provided, as shown in Illustration 1–4, for crated magnets.

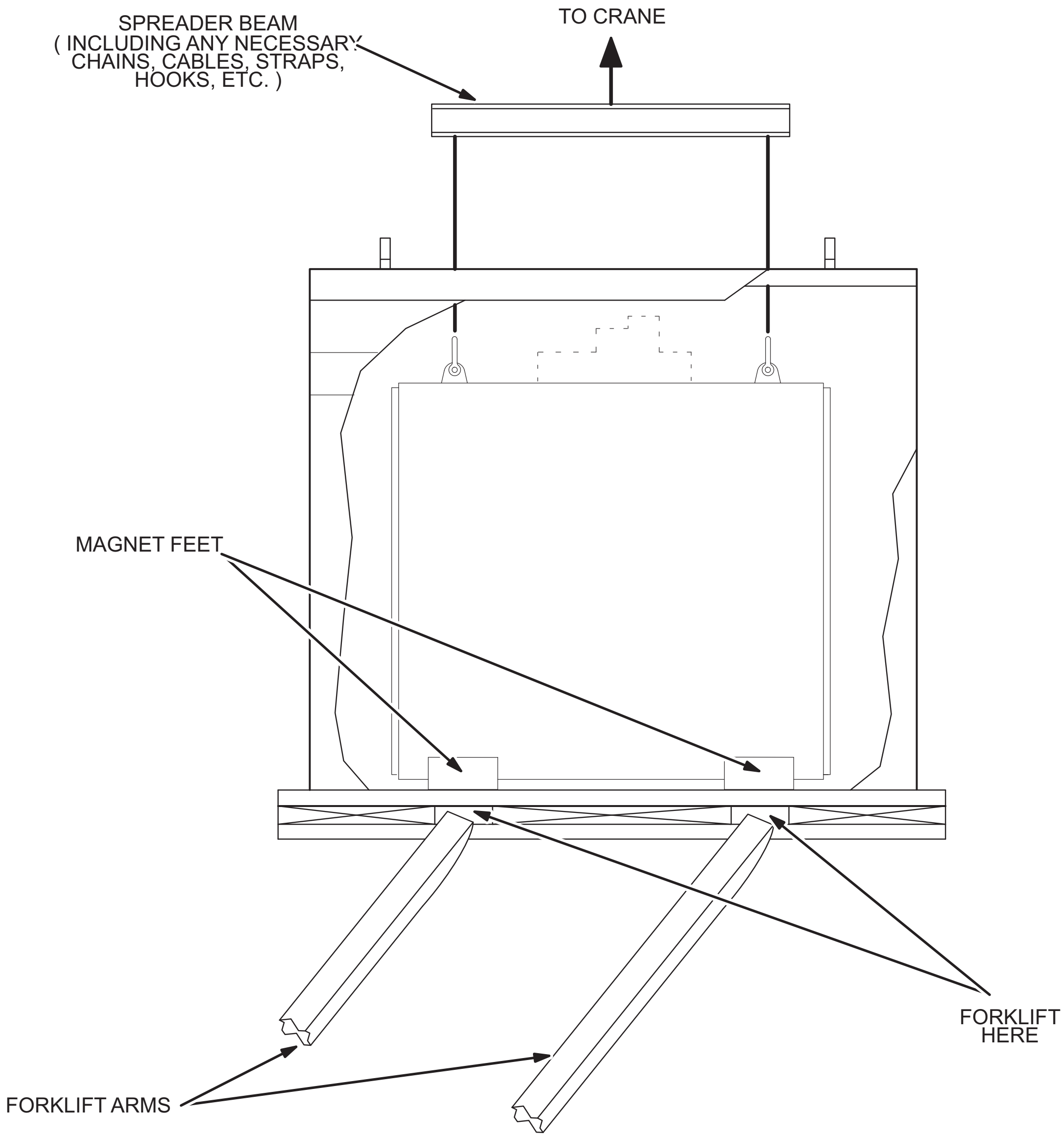


ACCESS PANEL, SHACKLE AND BRACKET LOCATIONS
ILLUSTRATION 1–4

1–3 UNLOADING / LOADING MAGNET AND CRATE (continued)

CAUTION

Extreme care must be exercised during forklift operations. The magnet crate must be picked up from the sides only. The forks MUST be placed directly under the four (4) feet of the magnet. The magnet feet can be identified by the steel plates attached to the pallet. Do not lift crate with straps fed through the pallet. This will crush the crate.



FORKLIFT / CRANE LIFTING OPERATIONS
ILLUSTRATION 1–5

1–3 UNLOADING / LOADING MAGNET AND CRATE (continued)

1. Position the Crane Hook centrally over the crated magnet to ensure a vertical lift force on the Lifting Brackets.

CAUTION

It is important to lift the magnet smoothly to avoid impact or jolts to the system which may damage the magnet.

2. Attach the rigging to the lifting shackles at each end of the crated magnet.

Note

Any combination of spreader beam, shackles and / or slings, 6 foot (1829 mm) minimum length, which can support a minimum of 18,000 pounds (8,163 Kg) may be used.

3. Lift crated magnet with crane to clear trailer.
4. Lower the crated magnet onto a flat surface.

1–4 PRE–DELIVERY INSTRUCTIONS

CAUTION

Make sure ALL site requirements / conditions, identified for the magnet in the site planning manual, are met before scheduling magnet delivery.

This will prevent installation delays, cryogen loss and resultant ongoing magnet quenches, potential damage, environmentally related problems and increased costs.

WARNING!

THE OXYGEN MONITOR MUST BE INSTALLED AND FUNCTIONAL BEFORE BRINGING THE MAGNET INTO THE EXAM ROOM. THIS MONITOR MUST GIVE AN AUDIO AND VISUAL WARNING WHEN THE OXYGEN LEVELS IN THE ROOM FALLS BELOW 18% TO NOTIFY PERSONNEL OF ANY POSSIBILITY OF ASPHYXIATION. REFER TO DIRECTION 15336 FOR OXYGEN MONITOR INSTALLATION INSTRUCTIONS.

1. Visit magnet site with rigging foreman before magnet delivery to plan the move.
2. Caution rigger that the magnet is extremely fragile. Sudden jolts can damage the magnet. Riggers aware of the cost of a magnet and its replacement usually use more care while handling magnet.
3. Make sure all roads and paths leading to exam room are level and free from obstacles and holes. Rigger will be required to construct platforms where needed.
4. If Roller Dollies are to be used, have rigger bring eight steel plates, 96in. x 24in. x 0.50 in. (2438mm x 610mm x 12.7mm), to place along delivery route.
5. Mark the magnet location on the floor before delivery. Refer to architectural drawings.

SECTION 2 – UNCRATING MAGNET SYSTEM

1. Remove all subsystem crates, except the Magnet Crate, from low–boy trailer using a crane or forklift. Inspect all crates for visible damage.

2. Move subsystem crates to a receiving location protected from the weather, preferably in proximity of the examination room and on the same level.

Note

Domestic shipments are made with a tarp covering on the magnet instead of a crate. See Illustration 2–1 for magnet weight (domestic and international shipements). For domestic shipments, skip to RIGGING, Section 3.

3. Inspect the crate containing the magnet and identify the crate side marked, “REMOVE THIS SIDE FIRST”. This designated side is to be removed first. See Illustration 2–1 for crate dimensions.

4. Position crane centrally over this designated crate side.

5. Connect crane to the two lifting rings located near the top of the crate side marked ”Remove This Side First” using a sling and shackles supplied by riggers. Tighten the sling snug. See Illustration 2–2.

6. Remove lag screws securing the face of the crate marked “REMOVE THIS SIDE FIRST” See Illustration 2–2.

7. Lift crate side with crane and clear from the area.

8. Secure four one ton (900 Kg) working load slings, minimum 6 foot length (1829mm), to four lifting rings assembled to the top corners of crate. Using one ton (900 Kg) anchor shackles.

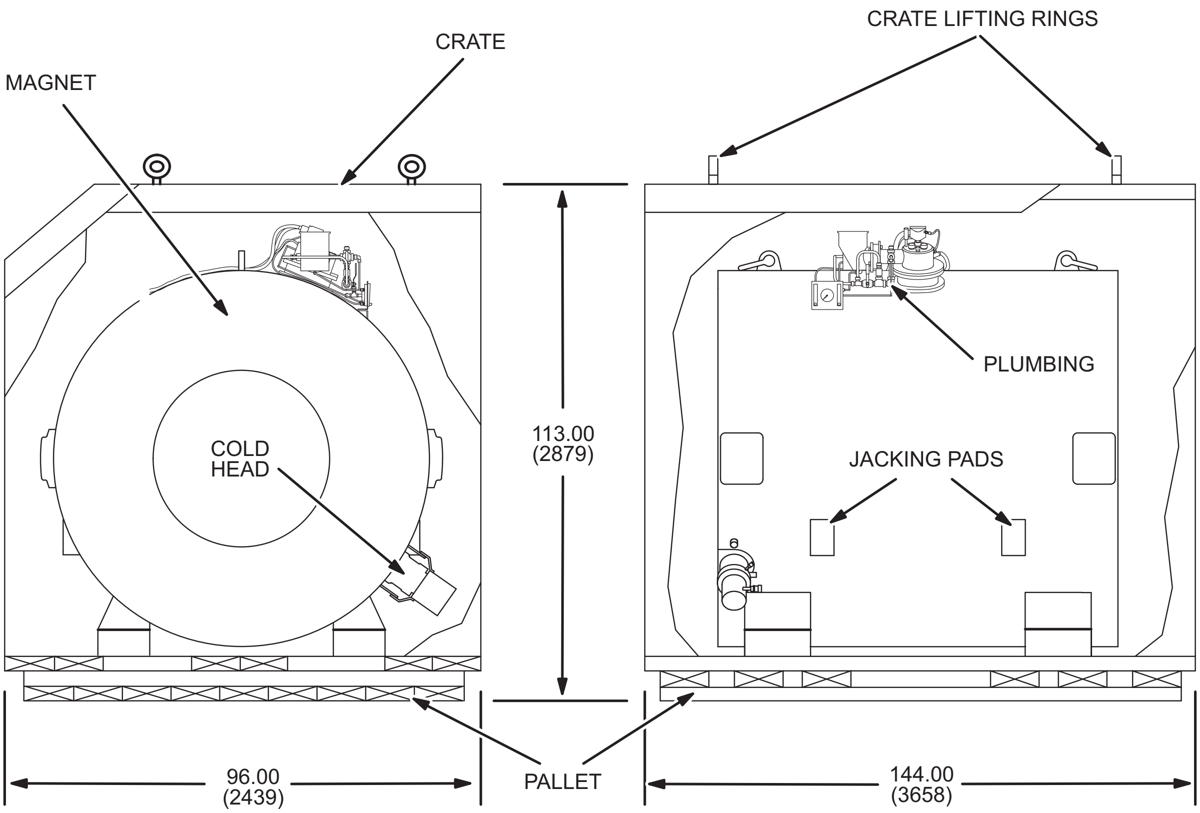
9. Position crane centrally over top section of the crate and attach slings to crane hook.

10. Remove lag screws only from the bottom portions of the three remaining crate sides. See Illustration 2–3.

11. Lift crate carefully using crane and clear crate from shipping skid containing magnet.

CAUTION

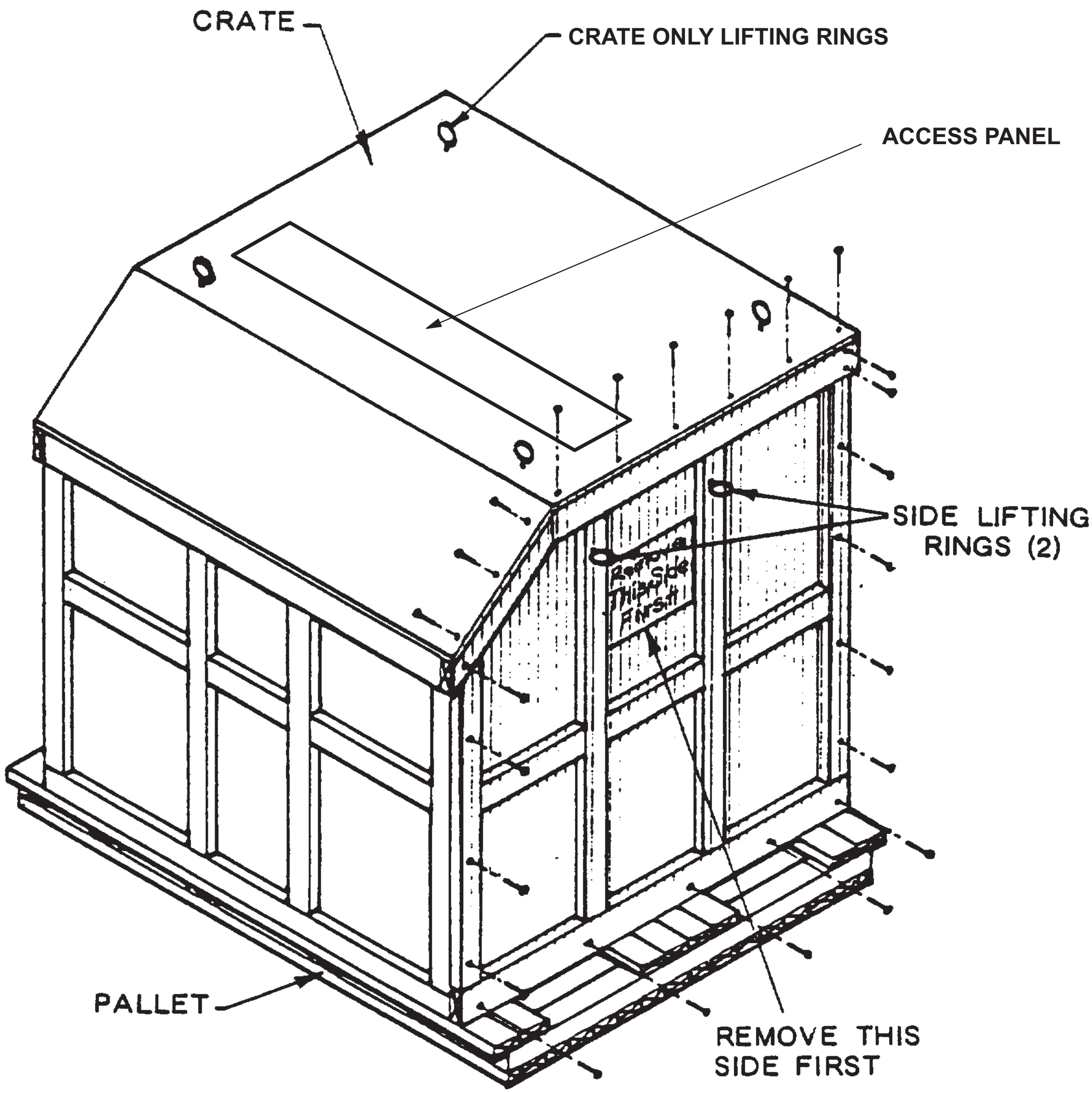
Make sure that the crate does not swing and hit the Magnet Cold Head, Vertical Penetration or Plumbing during the lifting process.



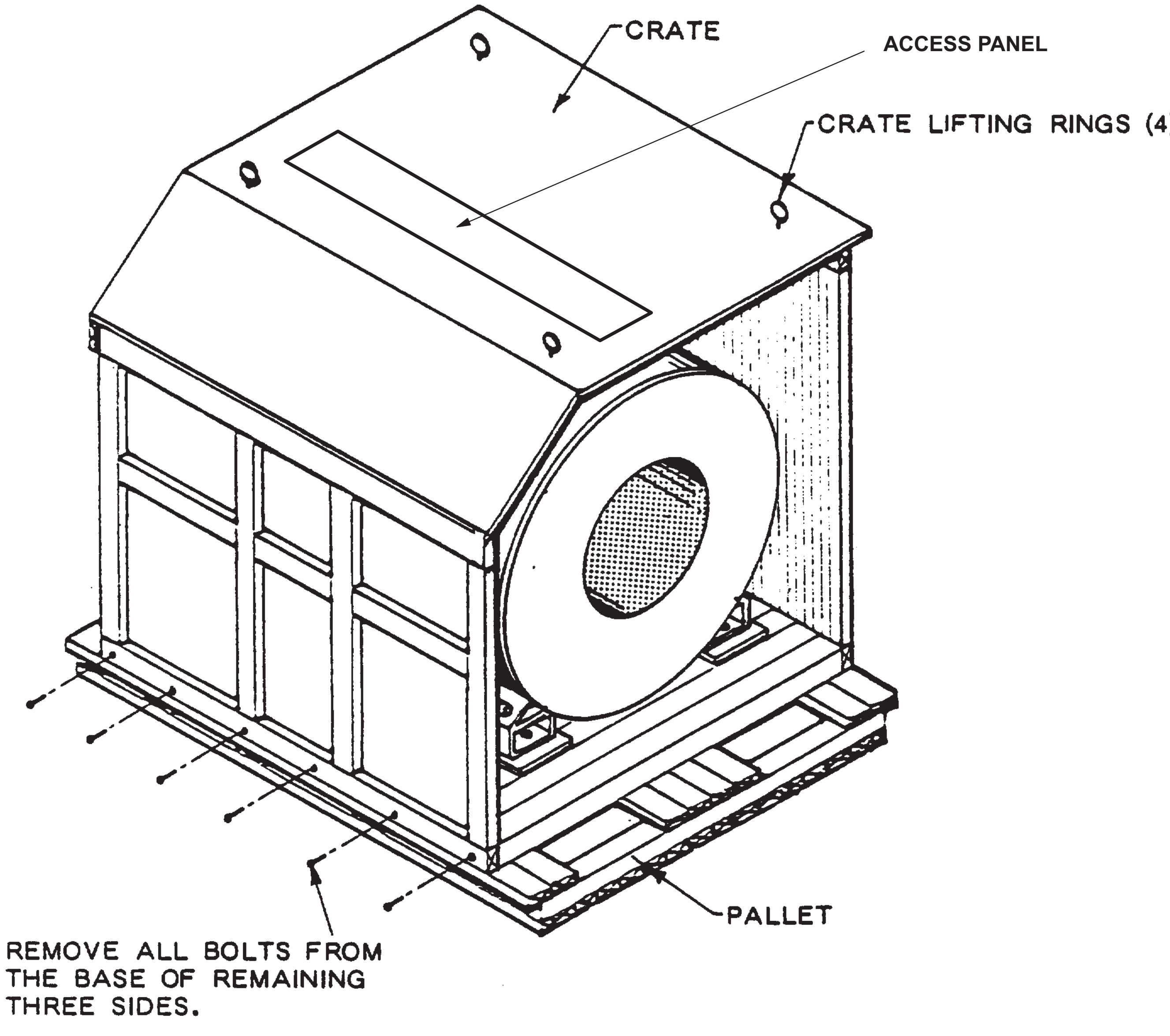
MAGNET CRATE
ILLUSTRATION 2–1

NOTE:

MAGNET LIFTING SHACKLES ARE LOCATED UNDER THE ACCESS PANEL DOOR



MAGNET CRATE FACE LAG SCREW REMOVAL
ILLUSTRATION 2–2



MAGNET CRATE BOTTOM LAG SCREW REMOVAL
ILLUSTRATION 2–3

SECTION 3 – MAGNET SYSTEM COMPONENT CHECK

Note

Shipment of magnet system components to the installation site may occur as a complete system or a drop shipment of major system components. Verify that all required magnet system components are present at the site to assure a complete and timely installation.

1. Locate the Pre–Delivery Information Package shipped with the magnet which contains the (Bill of Material) for the magnet system delivered. Check that all boxes indicated are present.
2. Check the contents of each box against its packing list when the boxes are brought into the MR Site.
3. Inspect the magnet for physical damage and icing/condensation on the body. If no problem found, unload magnet.

Note

Because of the higher Boiloff and helium gas flow through the vertical penetration, some frost on the Vertical Penetration is normal during periods when cold head has been shut off.

4. If icing or condensation present on the body or the bore of the magnet, check Liquid Helium level before unloading. See Set Up and Calibration, Section 1–6 for Cryogen Monitor set up.

Note

If magnet has been sitting for a period of time with the Cold Head inoperative, the Magnet may be depleted of Cryogens.

5. Refer to SET UP AND CALIBRATION, Section NO TAG for Cryostat Cooling /Filling Requirements. If pressure in vacuum cavity is greater than 1×10^{-6} Torr., contact MAC Team Leader before installing magnet.
6. Make sure the Shim Lead is “Engaged” in conformance with SET UP AND CALIBRATION, Section 6.

CAUTION

It is important to establish if any damage was sustained by the magnet or its system components during delivery or if any components are missing. This will result in the fast, proper follow–up of any problems and a timely installation.

7. Perform the Magnet Circuits Resistance Checks identified in Table 3–1, “MAGNET CIRCUITS RESISTANCE CHECK COLD (4.2 K)” and Table 3–2 “SHIM LEAD CHECKS”. If any problems are found the Shim Lead can be “retracted” to isolate the shorts from the shim coils.
8. Report any damage found in compliance with the “Damage In Transportation” note on the back side of the Service Manual Title Page.
9. Report all problems found to the Regional Magnet & Cryogenics (MAC) Team Leader. Report all missing components to the person identified on the Magnet Component Bill of Material.

TABLE 3–1
MAGNET CIRCUITS RESISTANCE CHECK
COLD (4.2K)

FUNCTION	CONNECTOR	PIN #	RESISTANCE (OHMS)	
			TYPICAL VALUES	MEASURED
MAIN COIL	MAIN COIL POWER LUGS OR J5–1	+ – 9,10	< 4 OHMS	
SUPERCONDUCTING SHIM COILS	CANNON (P1–A) AT MAGNET VERTICAL STACK			
Z1		1,19	0.5	
Z2		2,20		
Z3		3,21		
Z4		4,22		
Z5		5,23		
Z6		6,24		
C11+		16,19		
C11–		17,20		
S11+		9,19		
S11–		10,20		
C31		13,23		
S31		11,23		
SUPERCONDUCTING SWITCH HEATERS MAIN SWITCH	J 5–1 & J 5–2 ON MAGNET TERMINAL BOX (MS1–A3,A1)	1,2	22 – 27	
AXIAL SHIMS		5,6	27.5 – 32.5	
TRANSVERSE 1		7,6	9 –10	
TRANSVERSE 2		8,6	9 –10	

SECTION 4 – REMOVING THE MAGNET FROM THE SHIPPING SKID

CAUTION

The magnet must be moved as smoothly as possible at all times. If severe jolting is encountered, internal damage may result.

1. Unbolt and remove the four 1 inch bolts which secure the base frame to the magnet. For magnets used on Sierra systems (YMS), remove the 1.5 inch steel plates by removing 4 bolts on each foot. See Illustration 4–1.
2. Position the Crane Hook centrally over the magnet to ensure a vertical lift force on the Lifting Brackets.

CAUTION

It is important to lift the Magnet smoothly to avoid impact or jolts to the system which may damage the magnet. The magnet cannot be tipped by more than 30 degrees during any lifting operation using the lifting shackles/brackets.

3. Attach the rigging to the lifting shackles at each end of the magnet.

Note

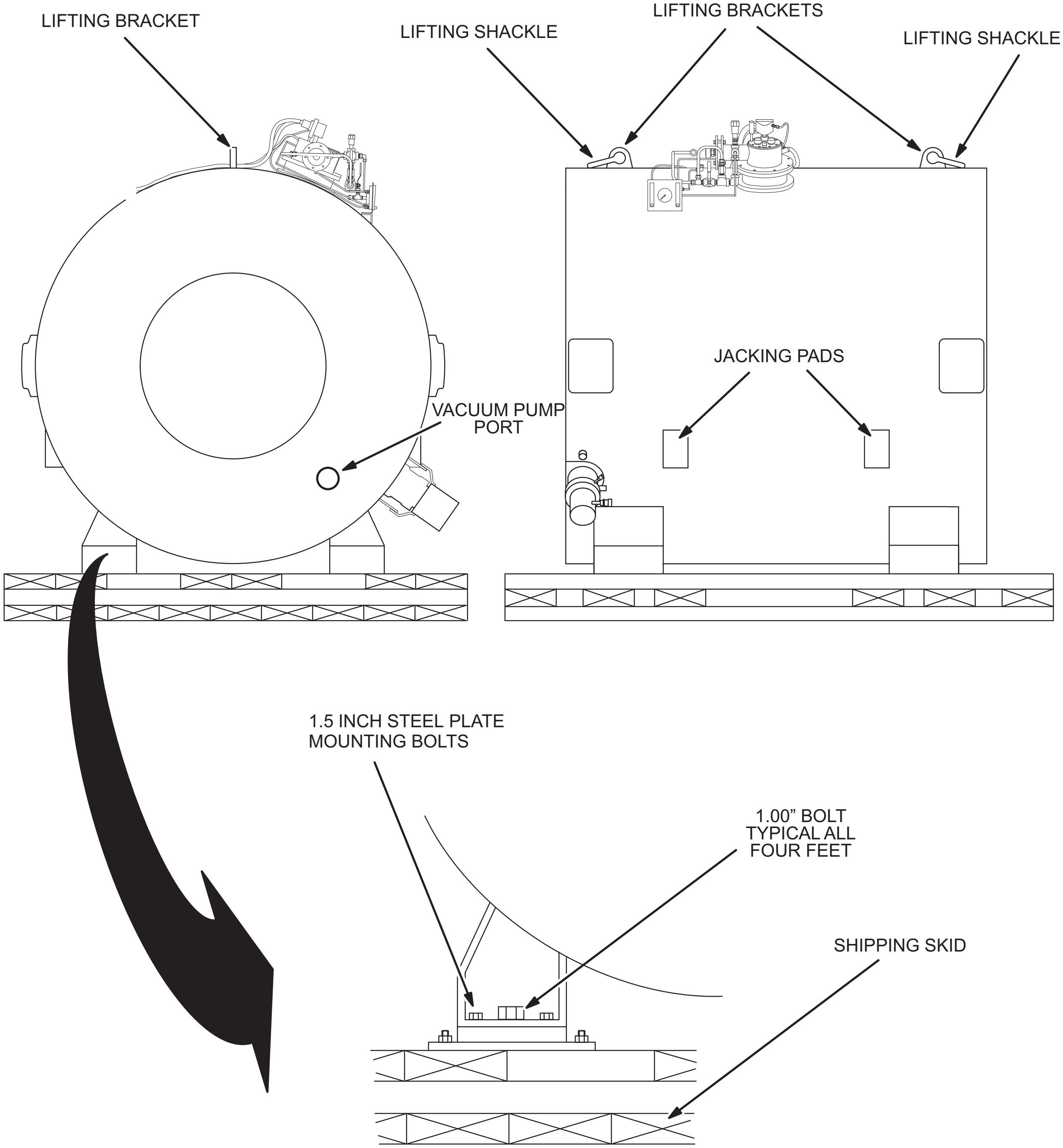
Any combination of spreader beam, shackles and/or slings, 6 foot (1829mm) minimum length, which can support a minimum of 15,500 pounds (7,029 Kg) may be used.

4. Lift magnet with crane to clear shipping skid and trailer.

CAUTION

When lowering the magnet, be careful not to shock the magnet which could result in over stressing the internal magnet support members.

5. Lower the magnet onto a flat surface.



MAGNET MOUNTING ON SHIPPING SKID
ILLUSTRATION 4–1

SECTION 5 – MOVING MAGNET TO EXAM ROOM

Description

Before moving magnet to exam room (fixed sites), make sure mounting holes, anchors and washers are installed as per Magnet Bolt Down Kit instructions. For normal installations use Magnet Bolt Down Kit R–Cat #R4391CA.

Note

Roller Dollies are recommended for moving magnet into the examination room as shown in Illustration 5–1. If Roller Dollies are used, place steel floor plates along the magnet delivery route.

CAUTION

Use the Jacking Pads located on the magnet to raise the magnet for Roller Dolly installation.

Use shims when rolling magnet with Roller Dolly over door thresholds and other inclines.

Note

The magnet location in the examination room must be marked. Use tape to mark the four corners of magnet on the examination room floor. Refer to architectural drawings to determine the exact location of magnet within the examination room.

1. Push magnet to exam room. If using a motorized tow vehicle, attach chains around the magnet base support pads and pull magnet.
2. If there are turns in the delivery route, adjust Roller Dollies to appropriate positions to negotiate turns.
3. Move Magnet into the Exam Room. If there is not enough vertical clearance for the Magnet, remove the 1.5 inch plates from the bottom of the magnet or the Vent Adapter from the Vertical Penetration to obtain additional clearance. If more clearance is still needed call the MAC Team Representative for further removal of Vertical Penetration components.
4. Position magnet on the examination room floor. If ceiling is in place, drop a plumb bob from the center of the vent hole and mark the floor at the location where the plumb bob points. This corresponds to the location of the center of the Vent Adapter on the Magnet.

Note

Table end of magnet (See Introduction, Illustration 4–1) must be oriented away from rear wall of exam room.

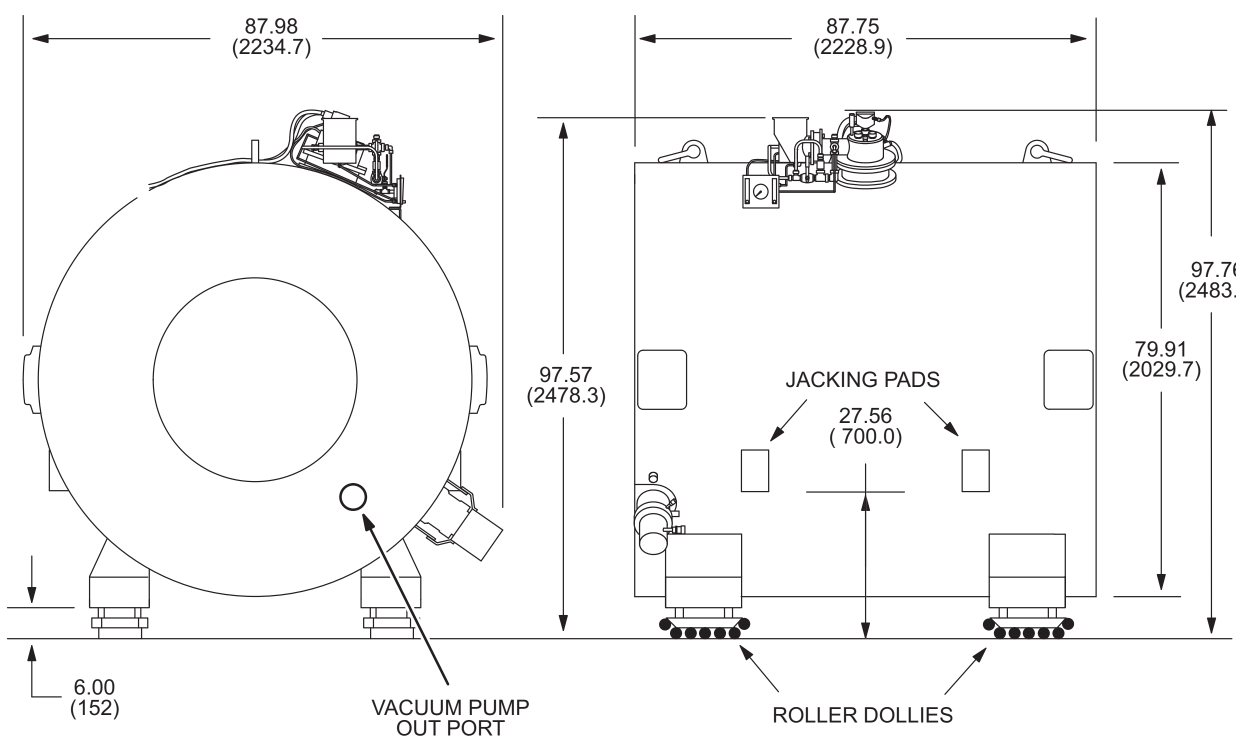
CAUTION

Keep magnet level at all times. Uneven jacking of corners could result in shifting of magnet on jacks.

5. Jack the magnet up sufficiently, at the 4 jacking pads, and remove the 4 Roller Dollies.

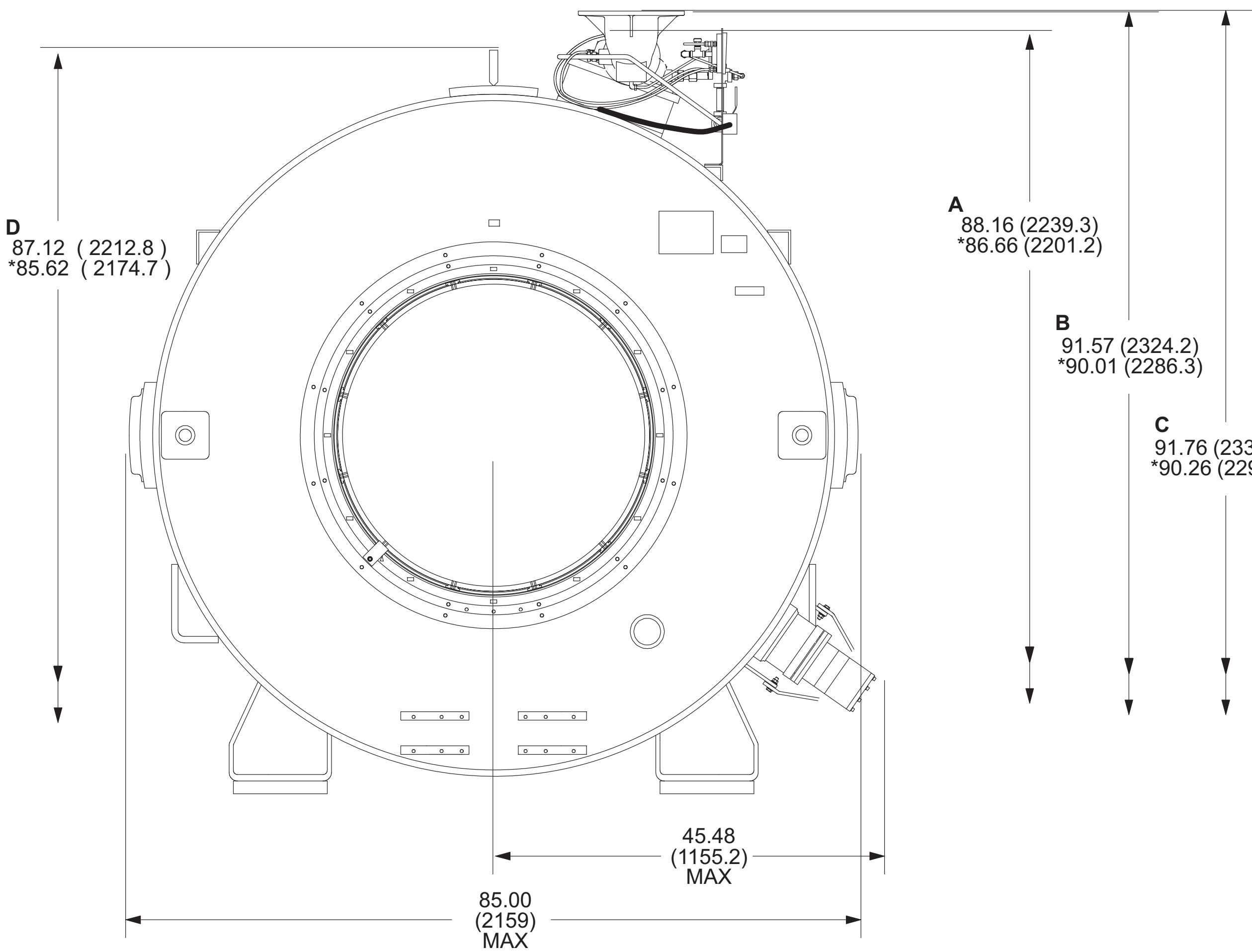
6. Slowly lower magnet onto the exam room floor.

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS)
* DIMENSIONS INCLUDE 1.50 (38.1) PLATES ON BOTTOM OF EACH FOOT.



MAGNET WITH ROLLER DOLLIES ATTACHED
ILLUSTRATION 5–1

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS)
* DIMENSIONS SHOWN WITHOUT FOOT BLOCKS ATTACHED TO MAGNET FEET.



A: DISTANCE TO UPPERMOST POINT ON TURRET COVER PLATE.
B: DISTANCE TO TOP OF VENT ADAPTER.
C: DISTANCE TO TOP OF SHIM LEAD CONNECTOR BOX.
D: DISTANCE TO TOP OF LIFTING LUG

Note

Minimum height for service clearance is 102.68 inches (2608mm). This clearance is needed for Ramp Lead, Shim Lead, and Fill Line installation.

MAGNET CLEARANCE DIMENSIONS
ILLUSTRATION 5–2

SECTION 6 – LEVELING MAGNET

1. Fill a 25 foot (7.6m) x 0 .5 inch (12.7mm) section of transparent tubing 3/4 full with water.
2. Position the ends of the tube adjacent to the 3mm deep grooves, located on the end flanges of the magnet, at the horizontal center. Assure that there is sufficient water in the tube so that the water level is above the groove height. See Illustration 6–1.
3. Measure the difference between the groove and water level height in the tube. There should be no more than 0.04 inch (1 mm) difference between any of the four measurements (four corners of the magnet).
4. If there is more than a 0.04 inch (1 mm) difference, jack up magnet and insert an aluminum shim plate under the appropriate corner of the Magnet Base Frame.

Note

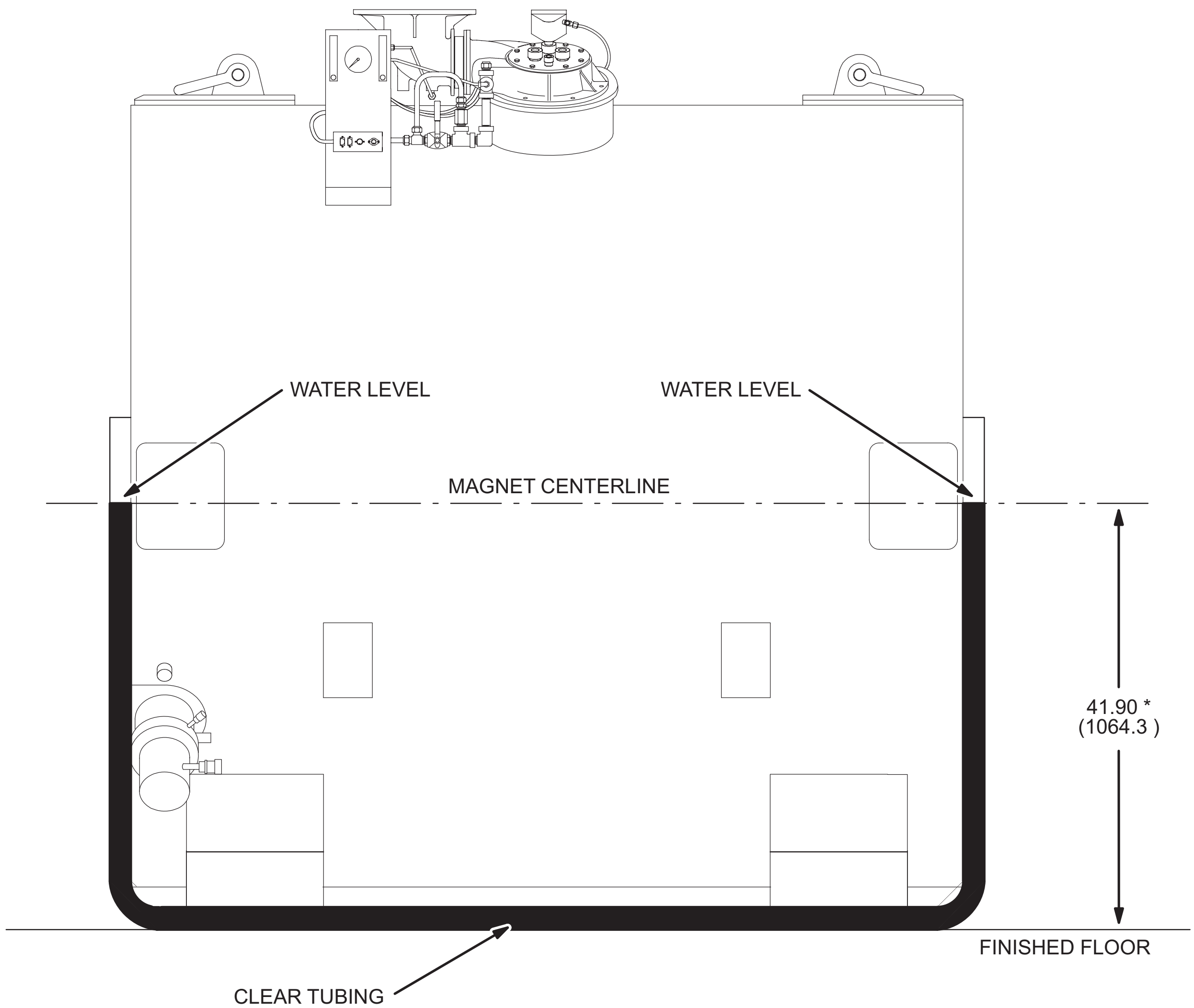
A Magnet Leveling Kit (46–260888G1), containing 12 – 0.062 inch (1.57mm) and 8 – 0.020 inch (0.5mm) shim plates, is shipped with the magnet.

5. Lower magnet onto shim plate and recheck the level.

Note

Repeat Steps 2 through 5 until the magnet is level (i.e., there is no more than a 0.04 inch (1mm) difference between any of the four measurements.

6. Allow magnet to settle on exam room floor for approximately 12 hours and recheck magnet level.
7. After 12 hours, bolt the magnet to the floor using the appropriate bolt down kit. If the magnet is in a Seismic Zone it will require the Magnet Bolt Down Installation Kit (P/N 2105959). All other SV installations require the Magnet Bolt Down Installation Kit (P/N 2104219) for reduced vibration sensitivity.



* MINIMUM DISTANCE BETWEEN MAGNET CENTERLINE AND FINISHED FLOOR IS 41.90 (1064.3) FOR FRONT AND REAR SHROUD CLEARANCE.

CHECKING MAGNET LEVEL
ILLUSTRATION 6–1